R15/R13

Code No: 125AP/115AP

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, March - 2021
COMPILER DESIGN
(R15 - Computer Science and Engineering; R13 - Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks

1.a) Write a LEX program to identify few reserved words of C language.
b) What is ambiguous grammar? How to eliminated left recursion from an ambiguous grammar? Explain. [7+8]

2. Eliminate the left-recursion from the following grammar:
$S \rightarrow (L) \mid a$
$L \rightarrow L,S \mid S$
Determine whether a predictive parser can be constructed for this grammar or not. Also construct the parsing table. [15]

3. Compute LR(0) items for the following grammar and construct SLR parsing table: [15]
$S \rightarrow L=R \mid R$
$L \rightarrow *R \mid id$
$R \rightarrow L$

4.a) Write a YACC program to evaluate arithmetic expressions.
b) Write an algorithm for constructing LALR Parser table. [7+8]

5. Explain briefly various data structures used to implement the symbol table. [15]

6. Discuss various loop optimization and load optimization techniques with an illustrative example. [15]

7.a) Write the algorithm for gencode( ) procedure.
b) Explain the following peephole optimization techniques: [7+8]
i) Algebraic Specification
ii) Strength Reduction

8. Explain the process of register allocation using graph coloring with an example. [15]

---ooOoo---

USED PAPER MARCH-2021